



UNIVERSITY OF THE PUNJAB

Roll No.

Fifth Semester 2017

Examination: B.S. 4 Years Programme

PAPER: Probability Theory (Theory)

Course Code: STAT-307

TIME ALLOWED: 30 mins.

MAX. MARKS: 10

Attempt this Paper on this Question Sheet only.

OBJECTIVE TYPE

Q.1 Tick the correct answer for the following multiple choice questions.

- I. The distribution having the moment generating function $\frac{1}{(3-2e^t)}$ can be classified as:
 - a. Negative binomial distribution
 - b. Geometric distribution
 - c. Binomial distribution
 - d. None of the above
- II. The probability is $p=0.80$ that a patient with a certain disease will be successfully treated with a new medical treatment. Suppose that the treatment is used on 40 patients. What is the "expected value" of the number of patients who are successfully treated?
 - a. 0.40
 - b. 20
 - c. 8
 - d. 32
- III. Suppose that the probability of event A is 0.2 and the probability of event B is 0.4. Also suppose that the two events are independent. Then $P(A/B)$ is:
 - a. $P(A) = 0.2$
 - b. $P(A)/P(B) = 0.2/0.4 = 1/2$
 - c. $P(A)P(B) = (0.2)(0.4) = 0.08$
 - d. None of the above
- IV. A coin is tossed three times. What is the probability that it lands on heads exactly one time?
 - a. 0.125
 - b. 0.250
 - c. 0.333
 - d. 0.375
- V. Whenever $P(\text{success}) < 1/2$ and the number of trials is fewer than ten, the shape of a given distribution will be:
 - a. Symmetrical but only if n is large
 - b. Positively skewed towards the left
 - c. Positively skewed towards the right
 - d. Symmetrical
- VI. The moment generating function of a random X is, $M_{x(t)} = \frac{2}{5} + \frac{1}{3}e^{2t} + \frac{4}{15}e^{3t}$. The expected value of X is

P.T.O.

- a. $22/15$
- b. $9/5$
- c. $17/15$
- d. $11/5$
- e.

VII. The probability that a leap year will have 53 Sundays is

- a. $1/7$
- b. $2/7$
- c. $2/53$
- d. $52/53$

VIII. In the outcomes of a discrete random variable follow a Negative binomial distribution, then their:

- a. Mean equals the variance of that discrete random variable
- b. Mean less than the variance of that discrete random variable
- c. Variance less than the mean of that discrete random variable.
- d. Both a and c

IX. Given $P(A) = 0.4$, $P(B) = 0.3$, $P(B/A) = 0.2$. What are $P(A \text{ and } B)$ and $P(A \text{ or } B) = ?$

- a. $P(A \text{ and } B) = 0.12$, $P(A \text{ or } B) = 0.58$
- b. $P(A \text{ and } B) = 0.08$, $P(A \text{ or } B) = 0.62$
- c. $P(A \text{ and } B) = 0.12$, $P(A \text{ or } B) = 0.62$
- d. $P(A \text{ and } B) = 0.08$, $P(A \text{ or } B) = 0.70$

25/72

X. Naveen is a high school basket ball player. She is a 70% free throw shooter. That means her probability of making a free throw is 0.70. What is the probability that Naveen makes her first free throw on her fifth shot?

- a. $5(0.70)(0.30)^4$
- b. $(0.70)(0.30)^4$
- c. $5(0.70)(0.30)^5$
- d. None of the above



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TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

SUBJECTIVE TYPE

Q.2 Write short answer on the following questions.

- a) Explain what is meant by a Random Experiment, a Sample Space and an Event.
- b) If $P(A) = 0.5$, $P(B) = 0.3$, and $P(A \text{ or } B) = 0.65$, are events A and B are independent?
- c) Explain how negative binomial distribution can be distinguished from binomial distribution.
- d) A coin is known to be unbalanced in such a way that heads only comes up 0.4 of the time.
 - (i) What is the probability that first head appears on 4th toss?
 - (ii) How many toss would it take, on average, to flip two heads?
- e) What is a geometric experiment and what are its properties?

26/72

(20)

- Q.3.a) Box A contains 5 green and 7 red balls. Box B contains 3 green, 3 red and 6 yellow balls. A box is selected at random and a ball is drawn at random from it. What is the probability that the ball drawn is green?
- b) There are four candidates for the office of the Highway Commissioner. The respective probabilities that they will be selected are 0.3, 0.2, 0.4, and 0.1 and the probabilities for a project approval are 0.35, 0.85, 0.45 and 0.15, depending on which of the 4 candidates are selected. What is the probability of the project getting approved?

(5+5)

Q.4 Compute mean, variance, coefficient of skewness and kurtosis of negative binomial distribution.

(10)

Q.5.a) Derive the Poisson distribution as the limiting form of the binomial distribution, stating clearly the assumptions you make.

- b) Three people each toss a coin and the odd man pays for the coffee. If the coins all show heads or all show tail, they are tossed again. What is the probability that a decision is reached in five tosses or fewer?

(5+5)